

1. Transport Layer & Data Link Layer

Given:

File size = 150 MB

Bandwidth = 50 Mbps

Segment size = 1500 bytes

Header + Trailer = 40 bytes/frame

$$\text{Segments} = \frac{150,000,000}{1500}$$

$$\text{segments} = 100,000$$

$$\begin{aligned}\text{Total Overhead} &= 100,000 \times 40 \\ &= 4,000,000 \text{ bytes} \\ &= 4 \text{ MB}\end{aligned}$$

$$\begin{aligned}\text{Transmission Time} &= \frac{1200}{50} \\ &= 24 \text{ sec}\end{aligned}$$

Result:

$$\text{segments} = 100,000$$

$$\text{Total Overhead} = 4 \text{ MB}$$

$$\text{Transmission Time} = 24 \text{ sec.}$$

2. Sliding Window

Given:

Window size = 6

Total frames = 48

Frame size = 1024 bytes

$$\text{Transmission Windows} = \frac{\text{Total frames}}{\text{Window size}}$$

$$= \frac{48}{6}$$

$$= 8$$

$$\text{Retransmissions} = 8 \times 1 = 8$$

One frame lost in every window

Result:

$$\text{Transmission Windows} = 8$$

$$\text{Retransmissions} = 8$$

→ Flow control prevents the sender from overwhelming the receiver, reducing congestion and improving efficiency.

3. IP Fragmentation

Given:

Packet = 12,000 bytes

MTU = 1500 bytes

$$\text{Header} = 20 \text{ bytes}$$

$$\text{Fragment} = 1500 - 20 = 1480 \text{ bytes}$$

$$\frac{12000}{1480} = 8.1$$

$$9 \times 20 = 180 \text{ bytes}$$

→ The Network Layer fragments large packets and reassembles them at the destination using identification numbers and fragment offsets.

4. Sliding Window.

$$\text{Transmission Windows} = \frac{48}{6} = 8$$

$$\text{Retransmissions} = 8 \times 1 = 8$$

→ ~~Regulates~~ transmission speed and avoids receiver buffer overflow.

5. Session Layer

Given:

$$\text{Video} = 600 \text{ MB}$$

Checkpoint every 60 MB

Failure after 390 MB

Checkpoints = 6

$390 - 360 = 30\text{MB} \rightarrow$ Retransmitted data.

Checkpoints before failure

60, 120, 180, 240, 300, 360.

$\rightarrow$ Synchronization checkpoints allow communication to resume from the last checkpoint instead of retransmitting the entire file.

6. Presentation Layer:

Given:

Original = 900MB

Compression = 40%.

Encryption Overhead = 5%.

Compressed size = $900 \times 0.60 = 540\text{MB}$

Encrypted size = $540 \times 1.05 = 567\text{MB}$

Reduction = $900 - 567 = 333\text{MB}$

Percentage = $\frac{333}{900} \times 100 = 37\%$

The presentation layer compresses data to reduce transmission size and encrypts it for confidentiality.

7: FTP Application:

Given:

$$\text{Data} = 30\text{KB}$$

$$\text{Throughput} = 120\text{Mbps}$$

$$\text{Request size} = 3\text{KB}$$

$$3\text{GB} = 3000\text{MB}$$

$$\text{Requests} = 3000/3 = 1000$$

$$3\text{GB} = 24000\text{MB}$$

$$\text{Transmission time} = 24000/120 = 200\text{sec}$$

→ The Application layer provides services like FTP, file transfer, authentication and user interaction

8. End - End Delay

Given:

$$\text{Application} = 4\text{ms}$$

$$\text{, propagation} = 18\text{ms}$$

$$\text{Transport} = 8\text{ms}$$

$$\text{, Transmission} = 12\text{ms}$$

$$\text{Network} = 5\text{ms}$$

$$\text{Data Link} = 2\text{ms}$$

$$\text{Physical} = 1\text{ms}$$

$$\text{Total} = 4 + 3 + 5 + 2 + 1 + 18 + 12 = 45 \text{ms}$$

$$\text{Total delay} = 45 \text{ms}$$

→ Each OSI layer performs specific processing before data is transmitted.

Protocol Overhead.

Given:

$$\text{Useful data} = 80 \text{MB}$$

$$\text{Frame size} = 1600 \text{ bytes}$$

$$\text{Overhead} = 80 \text{ bytes}$$

$$\text{Frames} = 80,000,000 / 1600 = 50,000$$

$$\text{Total Overhead} = 50,000 \times 80$$

$$= 4,000,000 \text{ bytes}$$

$$= 4 \text{MB}$$

Efficiency,

$$\text{Useful payload} = 1600 - 80 = 1520$$

$$\frac{1520}{1600} \times 100 = 95\%$$

10. Multiple OSI layers

Given:

Original file = 240MB

compression = 30%

segment size = 1400 bytes

Frame overhead = 60 bytes

Compressed size = $240 \times 0.70 = 168\text{MB}$

$$\text{Segments} = \frac{168,000,000}{1400} = 120,000$$

Protocol Overhead

$$= 120,000 \times 60 = 7,200,000 \text{ bytes}$$

$$= 7.2\text{MB}$$

$$\text{transmitted data} = 168 + 7.2 = 175.2\text{MB}$$